

Unveiling the Mysteries of Coral Bleaching:

Investigating Coral Bleaching Through Multiple Different Regression Methods

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Introduction

The objective of this study is to determine what covariates most heavily impact coral bleaching. We will use a data set that contains 2,394 observations spanning from 2006 to 2016 that is sourced from the Florida Reef Resilience Program (FRRP). Below we will discuss coral bleaching and several of the variables found in the data set.

Coral bleaching occurs when coral reefs are stressed by changes in conditions such as temperature, light, or nutrients, leading them to then expel the symbiotic algae living in their tissues, causing them to turn completely white. There are several factors that contribute to coral bleaching, including, but not limited to: rising sea temperatures, rising sea levels, and ocean acidification, all of which are consequences of climate change. This study will utilize data containing covariates such as exposure, distance to shore (meters), turbidity, cyclone frequency, depth (meters), wind speed (meters per hour), temperature (Kelvin), ClimSST, SSTA, and SSTA_DHW. Some other variables in the data set that ultimately will not be used are: site ID, sample ID, date (day, month, and year), degrees latitude, degrees longitude, and several different variations of temperature variables such as mean temperature (Celsius), minimum temperature (Celsius), maximum temperature (Celsius), TSA, and TSA_DHW (more explanation on this can be found in the Data Pre-processing and Exploration section). The response variable will be the percentage of coral bleaching occurring in transect segments, called Percent_Bleaching.

To enhance reader comprehension, we will further define a few of the variables mentioned above. These definitions are sourced from van Woesik and Kratochwill. Exposure refers to whether a site was exposed or not. A site was considered exposed if it had greater than 20 kilometers of fetch (the horizontal distance over which wave-generating winds blow). Otherwise, the site would be considered sheltered. Distance to shore is the distance of the sampling site from the nearest land in meters. Turbidity is the measure of relative clarity of a liquid. It is an optical characteristic of water and is a measurement of the amount of light that is scattered by material in the water ("Turbidity and Water"). Cyclone frequency refers to the number of cyclone events from 1964 to 2014. Depth is the depth of the sampling site in meters. ClimSST is the Climatological Sea-Surface Temperature (SST), which is based on weekly SSTs for the study time frame. SSTA is the Sea-Surface Temperature (SST) Anomaly, which is the weekly SST minus the weekly climatological SST. SSTA_DHW refers to the Sea Surface Temperature Degree Heating Weeks, which is the sum of the previous 12 weeks when the SSTA

is greater than or equal to 1 degree Celsius. TSA is the Thermal Stress Anomaly, which represents weekly SSTs minus the maximum of weekly climatological SSTs in degrees Celsius. TSA_DHW refers to the Thermal Stress Anomaly Degree Heating Weeks, which is the sum of the previous 12 weeks when the TSA is greater than or equal to 1 degree Celsius.

There are four main sections of data analysis and modeling for this study: (1) linear regression models, (2) nonparametric regression models, (3) regression tree models, and (4) stacked models. The linear regression section includes the following: multiple linear regression (both with interaction terms and without interaction terms), subset selection (best subset, forward stepwise, and backward stepwise), ridge regression, LASSO regression, principal components regression, and partial least squares regression. The nonparametric regression section includes the following: K-nearest neighbors, multivariate adaptive regression splines, a generalized additive model, local regression, and support vector regression. The regression tree section includes the following: a standard regression tree model, a random forest model, a bagged tree model, a boosted tree model, and a Bayesian additive regression tree model. The stacking section includes one model where the following three models are stacked: a generalized additive model, a random forest model, and a boosted tree model. This study will be carried out using the R programming language, which is a powerful tool designed for statistical computing and data visualization.

This study aims to find what covariates contribute to coral bleaching significantly, and what covariates do not. For the linear regression models, we will be able to determine what variables were ‘significant’ and what variables were not. This is equivalent to testing $H_0: \beta_i = 0$ versus $H_A: \beta_i \neq 0$, where $i = 1, \dots, p$ and p is the number of parameters in the model. In other words, we want to test whether the effect of a predictor, or interaction between predictors, is significant or not. We will also be looking at how well the models perform in terms of prediction using mean squared error. For the nonparametric models, we will look at things like optimal values for tuning parameters (such a K in K-nearest neighbors), the nonlinear effects of the predictors, and how well the models perform. For the regression tree models, we will evaluate variable importance (i.e. how much relative influence a variable had on the model) and how well the models perform. Lastly, for the stacked model, we will compare the three individual models used with the overall stacked model by looking at their prediction performance.

The data will be split into a training set and a testing set with 75% being allocated to the training set and the remaining 25% being allocated to the testing set. For many models, cross-validation will be used within the training set to tune the hyperparameters. All models' performance will be evaluated using the same testing data set. As will be seen in the upcoming data analysis sections, the regression tree models far outperformed the nonparametric models, and the nonparametric models far outperformed the linear regression models. The stacked model offered only a slight improvement over the random forest model, which was one of the best performing models. Overall, the stacked model, random forest model, bagged model, and boosted model were the four best performing models. Temperature-based variables such as temperature (Kelvin), ClimSST, SSTA, and SSTA_DHW were consistently among the most influential (or 'significant') predictors. Other variables such as wind speed and exposure were consistently deemed not 'significant' or among the least influential predictors. This makes sense, since the factors that cause coral bleaching (mentioned above) are direct consequences of climate change.

This report will start off with a section on data pre-processing and exploration, where we will investigate the predictor variables and determine which variables should or should not be included in the modeling. Next, there will be four sections for data analysis pertaining to the four different types of modeling: linear regression, nonparametric regression, regression trees, and stacking. The final section will be the conclusions of the study.

Data Pre-processing and Exploration

In this section, we will discuss the pre-processing and exploration of the data. The original data set contained 62 variables, including, but not limited to, site ID, sample ID, date (day, month, and year), site name, several variations of temperature variables using different metrics, and several variables dedicated to comments. For simplicity and usability of the data, we will remove all such variables. For our statistical models, we do not need variables such as sample ID or site ID, nor do we need 30 different variations of temperature variables using different metrics as they are likely to cause multicollinearity. Therefore, we have chosen to keep the following temperature variables: temperature (Kelvin), ClimSST, SSTA, SSTA_DHW, TSA, and TSA_DHW as these seem to be the most important temperature variables according to van

Woesik and Kratochwill. After removing unwanted and unnecessary variables, our data set contains the following predictors: exposure, distance to shore (meters), turbidity, cyclone frequency, depth (meters), wind speed (meters per hour), temperature (Kelvin), ClimSST, SSTA, SSTA_DHW, TSA, and TSA_DHW.

Next, we will investigate the correlation between the predictor variables. Figure 2 shows the correlation plot. We can see that temperature and TSA have a very high correlation amounting to 0.932. Similarly, SSTA_DHW and TSA_DHW are highly correlated with a correlation of 0.952. All other correlation values between predictors are acceptable and do not suggest strong correlation. Therefore, we will remove TSA and TSA_DHW from the data set since the temperature, SSTA, and SSTA_DHW variables convey the information we need. Our final data set for statistical analysis and modeling is a data set with the following 10 predictors: (1) exposure, (2) distance to shore, (3) turbidity, (4) cyclone frequency, (5) depth, (6) wind speed, (7) temperature, (8) ClimSST, (9) SSTA, and (10) SSTA_DHW.

As seen in Figure 1, the density of our response variable is heavily right skewed, with the majority of the values being between 0 and 40, roughly. The density resembles a log-normal or gamma distribution, taking on only positive values. Figures 5 and 6 show the densities of all the predictor variables. Figures 3 and 4 show scatter plots of each predictor variable versus the response variable. Visually, it appears that variables such as temperature, depth, SSTA, and SSTA_DHW influence the response. We will investigate these relationships further through statistical modeling using the models mentioned previously.

Data Analysis, Part I: Linear Regression Methods

In this section, we will discuss part one of the data analysis: linear regression. These linear models include multiple linear regression, subset selection, ridge regression, LASSO regression, principal components regression, and partial least squares regression. In order to use linear regression, there are some assumptions that must be met: (1) there exists a linear relationship between the predictors and the response, (2) the residuals are independent, (3) we have homoscedasticity, and (4) the residuals are normally distributed.

With most linear regression models, such as multiple linear regression, we can determine whether a variable is 'significant,' which is the equivalent to testing $H_0: \beta_i = 0$ versus $H_A: \beta_i \neq 0$.

0 for the i^{th} parameter. For other models, such as subset selection and LASSO regression, we can find which variables are 'significant' by simply looking at which predictors were included in the model. Other models such as PCR and PLS regression simply offer a reduction in dimensionality, and possibly overall test MSE.

The first multiple linear regression model, Model 1.1a, is a linear regression model with no interaction terms. The goal of multiple linear regression models is to predict the response variable based on a linear combination of two or more predictor variables by minimizing the sum of the squares of the residuals (known as least squares). Initially, this model contained all 10 predictor variables. After running this model, the variables wind speed, exposure, and ClimSST were determined not to be significant. All other variables were significant at a significance level of 0.01. The model was then run again without the variables that were not deemed significant. The test mean squared error (test MSE) of the model was found using the testing data set, and it was found to be 216.46. The estimated R^2 of the model was 0.0862. Figure 16 shows some plots of residuals as well as the QQ plot. We can see that there are possibly some outliers (seen in both plots of residuals), and the residuals versus fitted values plot has a pattern. A large portion of the values are centered in one area. This suggests that our model may not be a good fit.

The next multiple linear regression model, Model 1.1b, is a linear regression model with interaction terms. Initially, this model contained all 10 predictor variables and all possible combinations of interactions between them. Some of the variables in this model that were most important were ClimSST, turbidity, cyclone frequency, wind speed, and SSTA_DHW. Several interaction terms were also significant such as distance to shore with temperature, distance to shore with ClimSST, turbidity with depth, cyclone frequency with wind speed, temperature with SSTA_DHW, and SSTA_DHW with SSTA (note that if a variable was deemed significant in an interaction term, the original variable was also included in the model even if it was not significant). The test MSE of this model was found to be 206.77 and the estimated R^2 of the model was 0.1339, both of which are an improvement from the previous model. Figure 17 shows some plot of residuals as well as the QQ plot. We can see that we most likely have some outliers (residuals verses leverage plot), but the residuals versus fitted values plot is better behaved than the previous model. Our QQ plot also achieves a straighter line in this model compared to the previous model. Therefore, we can conclude that this linear model with interaction terms is an improvement on the previous model that did not have interaction terms.

The next part of this linear regression section includes three model selection methods: (1) best subset selection, (2) forward stepwise selection, and (3) backward stepwise selection. All three of these methods chose the exact same model: a model with 7 predictor variables including distance to shore, turbidity, cyclone frequency, depth, temperature, SSTA, and SSTA_DHW. The variables wind speed, exposure, and ClimSST were not included. These 7 variables are the same variables that were deemed significant in our original multiple linear regression model with no interaction terms.

The next linear model, Model 1.3, is a ridge regression model with λ chosen by cross-validation. Ridge regression is a form of regularization, which is a statistical method to reduce errors caused by overfitting on training data. Ridge regression specifically corrects for multicollinearity in regression analysis (Murel and Kavlakoglu). The hyperparameter λ controls the weighting of the penalty term. In ridge regression, the coefficients of the variables are shrunk towards 0, but are never actually shrunk to be exactly 0. For this model, our optimal value for λ was found to be 0.327. Using the ‘one standard error rule,’ our optimal λ value was found to be 68.926. Many of the variables in this model were shrunk to be very small, such as distance to shore, cyclone frequency, depth, wind speed, ClimSST, and SSTA_DHW. The variables turbidity, cyclone frequency, depth, wind speed, and exposure all had negative effects on the linear model. The test MSE of this model was found to be 228.15. Therefore, this model performed worse than the two previous multiple linear regression models.

The next linear model, Model 1.4, is a LASSO regression model with λ chosen by cross-validation. The hyperparameter λ controls the weighting of the penalty term. LASSO regression is similar to ridge regression, except that LASSO regression will shrink coefficients to be exactly 0, meaning they are not included in the model. For this model, our optimal value for λ was found to be 0.01. Using the ‘one standard error rule,’ our optimal λ value was found to be 2.105. The only predictor variables with non-zero coefficients were temperature and SSTA_DHW. The test MSE of this model was found to be 226.19. Therefore, this model performs worse than the previous three models.

The next linear model, Model 1.5, is a principal components regression model (PCR) with the tuning parameter, M (the number of components), chosen by cross-validation. Principal components regression is a statistical technique that combines principal components analysis (PCA) with linear regression. PCA is a mathematical algorithm that reduces the dimensionality

of the data while keeping most of the variation, depending on how many ‘components’ are used. Principal components accomplish this reduction in dimensionality by identifying ‘directions’ of variance within the data, i.e. the first component would ‘point’ in the direction where the variation of the data is largest, the second component would ‘point’ in the direction where the variation of the data is second largest, and so on. For this model, we did not have 100% of our variance explained until we reached all $M = 10$ components, which is also where the lowest cross-validation error occurred. This is usually the case for PCR. However, to combat having to include all components, we can say that we would like to have at least 85% of our variance explained by the model. When the model reached $M = 7$ components, we attained approximately 90% of our variance explained. Using 7 components, the test MSE of this model was found to be 225.13. Therefore, this model is an improvement on the ridge and LASSO regression models, but it is not an improvement on the two multiple linear regression models.

The last linear model, Model 1.6, is a partial least squares regression model (PLS) with the tuning parameter, M , chosen by cross-validation. A drawback of PCR is that it does not consider the response variable when calculating the components. PLS is a technique that is related to PCR, however, with PLS regression the model attempts to find linear combinations that explain the variation in *both* the response variable and the predictor variables, as opposed to PCR, which just focuses on the predictor variables. For this model, we (again) did not have 100% of our variance explained until we reached all $M = 10$ components. The lowest cross-validation error was 15.44 and was achieved with $M = 6$ components. However, we did not have at least 85% of our variance explained until we reached $M = 8$ components, with approximately 89% of our variance explained. Using 8 components, the test MSE of this model was found to be 216.76. This is an improvement on the ridge, LASSO, and PCR models. This model has nearly the same MSE as the original multiple linear regression model with no interaction terms.

In conclusion, the two multiple linear regression models as well as the partial least squares regression model performed the best out of all the linear models. Figure 8 shows a bar plot comparing all test MSE values for all linear models. Figure 9 shows a table comparing the test MSE as well as test RMSE (root mean squared error) values for all linear models.

Data Analysis, Part II: Nonparametric Regression Methods

In this section, we will discuss part two of the data analysis: nonparametric regression. These nonparametric models include K-nearest neighbors, multivariate adaptive regression splines, a generalized additive model, local regression, and support vector regression. Nonparametric models are highly desirable in some cases because they require no assumptions about the population probability distribution. These models do not assume a specific functional form, which makes them versatile in capturing complex patterns in data. The number of parameters in a nonparametric model may vary with the size of the data, which can lead to increased computational complexity, but greater flexibility (“Difference between Parametric and Non-Parametric Models in Machine Learning”). Nonparametric models offer many ways to determine what variables are important. For example, generalized additive models show us which predictors had strong nonlinear relationships with the response variable. Multivariate adaptive regression splines models show us how the model behaved with differing degrees and number of components, which can be useful for determining if the relationships are, in fact, nonlinear.

The first nonparametric model, Model 2.1, is a K-nearest neighbors (KNN) regression model with K chosen by cross-validation. This model can be fitted in R using the `caret` package. KNN regression is a regression method where the target value for a ‘new’ data point is predicted based upon the average of the target values of its K-nearest neighbors in the training set. This is why we must choose K with cross-validation: to find out how many “neighbors” are optimal in order to attain the lowest MSE. For this model, our optimal K value was found to be 97 (see Figure 20). Once the optimal value for K was found, the model was refitted with this value. The test MSE of this model was found to be 237.23.

The next nonparametric model, Model 2.2, is a multivariate adaptive regression splines (MARS) model. MARS is an adaptation of CART (classification and regression trees) that allows for additive terms to be entered into the model. It can be applied to both classification and regression problems. The MARS algorithm is an extension of linear models that makes no assumptions about the relationship between the response variable and the predictor variables. In other words, it is a nonparametric regression method that builds multiple linear regression models across the range of predictor values (“Mars: Multivariate Adaptive Regression Splines:

Intro”). This model can be fitted in R using the `earth` package. Figure 21 shows the plot of the MARS fit. Our optimal model was a model with a degree of 2 and 18 terms. The test MSE of this model was found to be 175.95. This is the lowest MSE we have achieved so far using any model.

The next nonparametric model, Model 2.3, is a generalized additive model (GAM). Generalized additive models provide a general framework for extending a standard linear model by allowing non-linear functions of each of the variables, while maintaining additivity (Maity, Arnab). This model can be fitted in R using the `gam` package. Figures 22 and 23 show the fit of the GAM model for each predictor variable. An ANOVA table was also constructed when the model was built, indicating very strong nonlinear relationships between the response variable and ClimSST as well as SSTA_DHW, and a moderately strong linear relationship between the response and temperature. The following variables were deemed significant at a significance level of 0.01: wind speed, temperature, ClimSST, SSTA, and SSTA_DHW. The test MSE of this model was found to be 182.47. This is an improvement on the KNN model, Model 2.1, but it is not an improvement on the MARS model, Model 2.2.

The next nonparametric model, Model 2.4, is a local regression model. Local regression refers to a class of regression techniques that achieve flexibility in estimating the regression function by fitting different, but simple, models separately at each target point (Maity, Arnab). There are two tuning parameters for a local regression model: span and degree. The span determines the number of points used for smoothing. A larger value for the span results in a smoother function (more points used for smoothing), whereas a smaller value for the span results in a less smooth function (less points used for smoothing). For this model, the optimal values for span and degree were 0.5 and 1, respectively. The model was then refit with these optimal values. The test MSE of this model was found to be 182.85. This is nearly identical to that of the GAM model, Model 2.3, but not quite as good as that of the MARS model, Model 2.2.

The last nonparametric model, Model 2.5, is a support vector regression (SVR) model. This model can be fitted in R using the `e1071` package. The goal of SVR is to find a function that approximates the relationship between the predictor variables and the response variable, while minimizing the prediction error. There are two important hyperparameters in an SVR model, ϵ (epsilon) and cost. In the model, epsilon specifies the epsilon-tube, within which no penalty is associated in the training loss function with points predicted within a certain distance (epsilon) from the actual value (“SVR”). It must be non-negative. Cost is a regularization

parameter, where the strength of the regularization is inversely proportion to cost, i.e. if we have smaller values of cost, the regularization is stronger, and vice versa. Our SVR model was fitted with an epsilon of 0.1 and a cost of 1, with a “radial basis kernel,” which is a kernel that captures nonlinear features. The test MSE of this model was found to be 182.42. This is nearly identical to that of the GAM and local regression models, Models 2.3 and 2.4.

In conclusion, the MARS model was the best model, with a test MSE of 175.95. The GAM, local regression, and SVR models followed closely with test MSE values of 182.47, 182.85, and 182.42, respectively. Figure 10 shows a bar plot comparing all test MSE values for all nonparametric models. Figure 11 shows a table comparing the test MSE as well as test RMSE values for all nonparametric models.

Data Analysis, Part III: Regression Tree Methods

In this section, we will discuss part three of the data analysis: regression trees. These regression tree models include a standard regression tree, a random forest model, a bagged tree model, a boosted tree model, and a Bayesian additive regression tree model. Regression trees are a subset of nonparametric models, which therefore means they have little to no assumptions required to implement them, as discussed in the previous section. Regression tree models allow us to determine which predictor variables are most influential in a model, i.e. which variables are ‘significant.’ Many different models constructed in R are accompanied with bar plots demonstrating which variables had the most influence, and similarly, which variables had the least influence.

Before moving on in this section, we will describe the ideas behind bagged and boosted tree models. Bagging (also known as bootstrap aggregation) is when a random sample of the data set is chosen with replacement, meaning that the individual observations can be chosen more than once. After generating several data samples, these weak models are then trained independently (“What Is Bagging?”). The predictions of these different weak models are then averaged to get the final prediction. Boosting works in a similar way to bagging, except that the trees are grown sequentially, meaning each tree is grown using information from previously grown trees. Boosting does not involve bootstrap sampling; instead, each tree is fit on a modified

version of the original data set (Maity, Arnab). Basically, each new tree attempts to fix mistakes in the previous tree. The final ‘boosted’ model is then used for prediction.

The first regression tree model, Model 3.1, is a standard regression (decision) tree model. This model can be fitted in R using the `rpart` package. Figure 24 shows a plot of the regression tree. SSTA_DHW was used to make the initial split. The next two splits were determined by temperature. Almost all the remaining splits were determined by temperature, SSTA, or SSTA_DHW. There were a few splits that were determined by other variables such as distance to shore, cyclone frequency, and ClimSST. Overall, temperature, SSTA, and SSTA_DHW have the most influence on this regression tree, i.e. they are the most ‘important’ predictors. The test MSE of this model was found to be 151.27, which is the lowest MSE achieved thus far.

The next regression tree model, Model 3.2, is a random forest model. This model can be fitted in R using the `randomForest` package. The random forest algorithm is a powerful tree learning technique. Random forests offer an improvement over bagging by decorrelating the trees (Maity, Arnab). When bagging is used, there is sometimes a very ‘strong’ predictor that always determines the first split of the tree. This causes all the bagged trees to look quite similar, and the predictions from these trees will be highly correlated. To overcome this problem, random forest models force each split to consider only a subset of predictors (Maity, Arnab). Thus, we drastically reduce the correlation between the trees in the model.

In our random forest model, the number of trees that produced the lowest MSE was 499 trees (see Figure 25). The most influential predictor was SSTA_DHW (see Figure 26). The following most influential predictors were temperature, ClimSST, SSTA, distance to shore, depth, turbidity, cyclone frequency, wind speed, and exposure, in that order. Meaning that variables such as exposure and wind speed were among the least influential predictors. The test MSE of this model was found to be 121.88, which drastically improves on all other models motioned thus far.

The next regression tree model, Model 3.3, is a bagged tree model. The number of bootstrap samples used was 150, the minimum number of observations required in a node to split was 2, and the ‘complexity parameter’ was set to 0, meaning that we don’t require the model to be able to improve the overall fit by any amount to perform a split. When we set the minimum number of observations required to split and the complexity parameter to very small numbers, we essentially allow the trees to grow extremely deep, which leads to trees with high variance,

but low bias. Then when we apply bagging, we're able to reduce the variance of the final model while keeping the bias low. The influential predictors here are in quite a different order compared to the random forest model. Our most influential predictor was distance to shore, followed by depth, turbidity, cyclone frequency, temperature, ClimSST, SSTA, wind speed, SSTA_DHW, and exposure, in that order (see Figure 27). Once again, exposure and wind speed were among the least influential predictors. The test MSE of this model was found to be 120.44, which improves slightly on the previous random forest model, Model 3.2.

The next regression tree model, Model 3.4, is a boosted tree model. This model can be fitted in R using the `gbm` package. When running this model, we must specify the number of trees for R to use. In this case, we specified 5000 trees. The most influential predictor was SSTA_DHW, followed by distance to shore, ClimSST, SSTA, temperature, depth, turbidity, cyclone frequency, wind speed, and exposure, in that order (see Figures 28 and 29). Once again, wind speed and exposure are among the least influential predictors. The test MSE of this model was found to be 155.51, which is not as good as the other regression tree models we have done thus far.

The last regression tree model, Model 3.5, is a Bayesian additive regression tree (BART) model. This model can be fitted in R using either the `BART` or `bartMachine` package. Bayesian additive regression trees, like other models discussed thus far, rely on previously generated regression trees. In BART methodology, aspects from bagging and boosting are used, such as the random manner in which the new trees are generated (bagging and random forests) as well as each tree trying to 'fix' the previous tree (boosting), i.e. account for some aspect in the data that has not been accounted for yet. This BART model had 100 trees, 1,000 posterior samples, and a burn-in of 250 (in Bayesian statistics and Monte Carlo simulations, the first few 'draws' are discarded, so that the algorithm has a better chance of reaching the high-probability region of the posterior distribution). The test MSE of this model was found to be 143.22, which is better than the standard regression tree and boosting but is not better than the random forest and bagged models.

In conclusion, the random forest and bagged models were the best models, with test MSE values of 121.88 and 120.44, respectively. The BART model performed well, with a test MSE of 143.22. The standard regression tree and boosted models had the lowest performance, with test MSE values of 151.27 and 155.51, respectively. Figure 12 shows a bar plot comparing all test

MSE values for all regression tree models. Figure 13 shows a table comparing the test MSE as well as test RMSE values for all regression tree models.

Data Analysis, Part IV: Stacking

In this section, we will discuss part four of the data analysis: stacking. Stacking involves training a new learning algorithm to combine the predictions of several base learners (Maity, Arnab). Stacking in R can be done using the `h2o` package. There is only one stacked model in this study, Model 4.1. In this stacked model, a generalized additive model, a random forest model, and a boosted tree model are stacked together, and their predictions are combined. We must code each model individually using the following functions from the `h2o` package:

`h2o.gam()`, `h2o.randomForest()`, and `h2o.gbm()`. We then use the function `h2o.stackedEnsemble()` to stack the three models together. Once this is done, we are able to use the model for prediction. Figure 30 shows the correlations between the predictions from each model used. We can see that the predictions from the random forest model and boosted model are highly correlated, with a correlation of 0.949. The test MSE values of the three individual models included were: 227.67 for the generalized additive model, 118.25 for the random forest model, and 124.16 for the boosted tree model. The test MSE of the overall stacked model was 117.23. This makes the stacked model the overall best model in this study.

In conclusion, the stacked model offered a slight advantage over the random forest model. Figure 14 shows a bar plot comparing all test MSE values for the three individual models as well as the stacked model. Figure 15 shows a table comparing the test MSE as well as test RMSE values for the three individual models as well as the stacked model.

Conclusion

All models' performance was evaluated using the same testing data set. The regression tree models far outperformed the nonparametric models, and the nonparametric models far outperformed the linear regression models. The stacked model was the best overall but offered only a slight improvement over the random forest model. Overall, the stacked model, random forest model, bagged model, and boosted model were the four best performing models.

It is possible that the linear regression models performed poorly compared to other models because some or all of our assumptions in order to use linear regression were violated, such as the normality assumption. We know from the Data Pre-processing and Exploration section that our response variable resembles a gamma or log-normal distribution as opposed to a normal distribution. The distributions of any of the predictor variables also did not resemble a normal distribution. Also, it is quite possible that the relationship between the response variable and the predictor variables is simply not linear. This could explain why the nonparametric models (including regression trees) performed so much better. Nonparametric models, such as multivariate adaptive regression splines or regression tree models, are able to capture nonlinear relationships in the data, making them more desirable when we are faced with a problem where relationships between the predictor variables and the response variable are not necessarily clear.

In conclusion, the objective of this study was to determine what covariates most heavily impact coral bleaching. There are several factors that contribute to coral bleaching, including, but not limited to: rising sea temperatures, rising sea levels, and ocean acidification. Temperature-based variables such as temperature (Kelvin), ClimSST, SSTA, and SSTA_DHW were consistently among the most important (or ‘significant’) predictors. This makes sense in the context of the problem because many factors, like the ones listed above that are known to impact coral bleaching, are direct consequences of climate change. In the generalized additive model, Model 2.3, there were very strong nonlinear relationships between the response variable and ClimSST as well as SSTA_DHW, and a moderately strong linear relationship between the response and temperature. In the regression tree models, temperature, ClimSST, SSTA, and SSTA_DHW were commonly among the most influential variables. Other variables, such as wind speed and exposure, were consistently deemed ‘insignificant’ or were among the least influential variables. All the above provide evidence for what was previously stated: coral bleaching is a direct consequence of climate change.

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Appendix

Figure 1 (below): Density of response variable

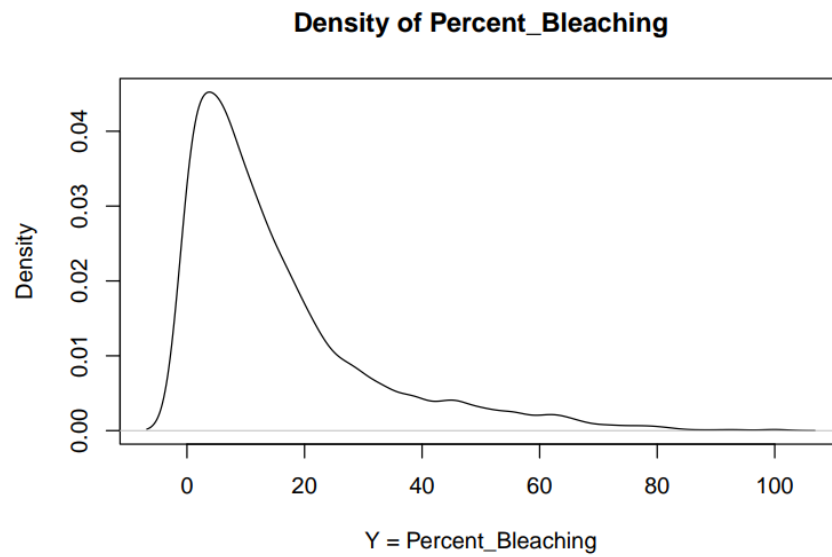


Figure 2 (below): Correlation plot of predictor variables

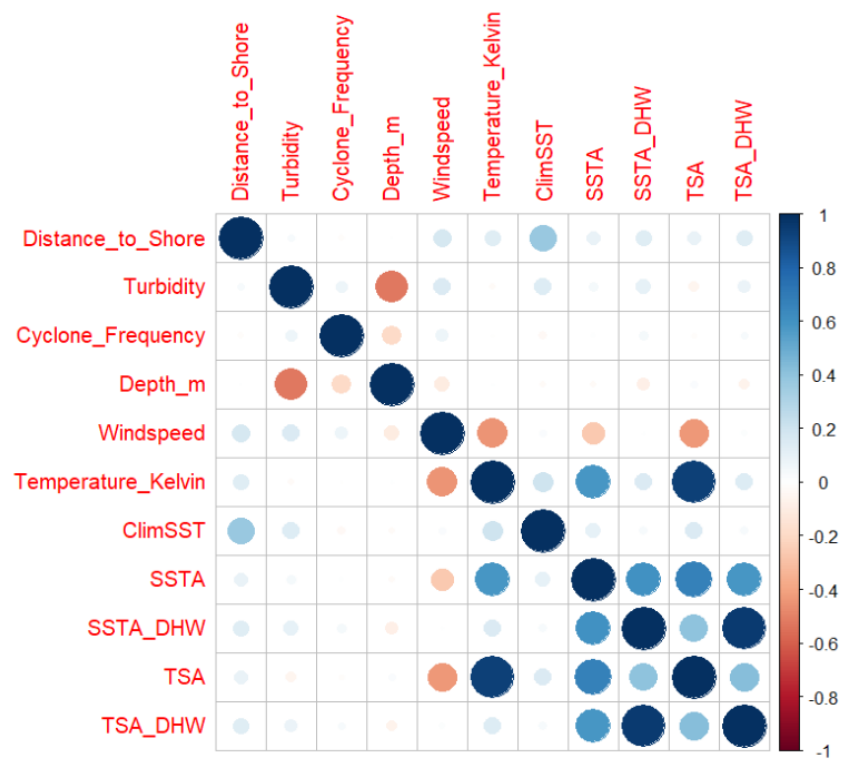


Figure 3 (below): Scatter plots of predictors vs response

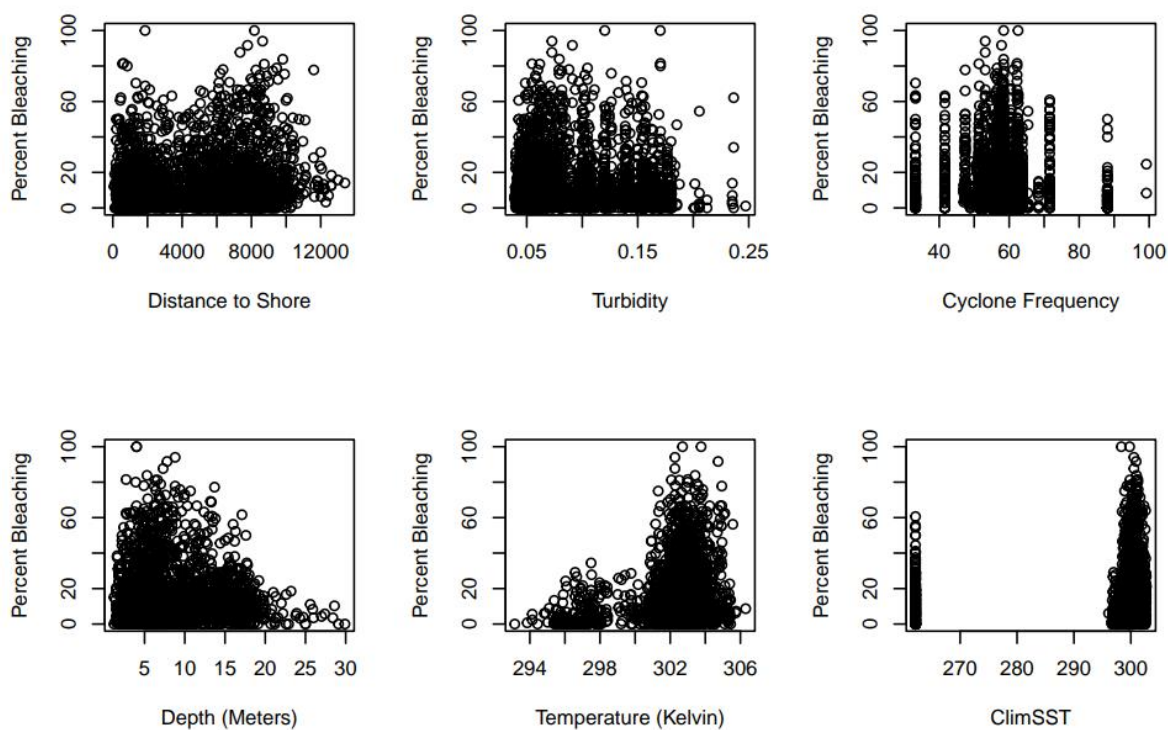


Figure 4 (below): Scatter plots of predictors vs response

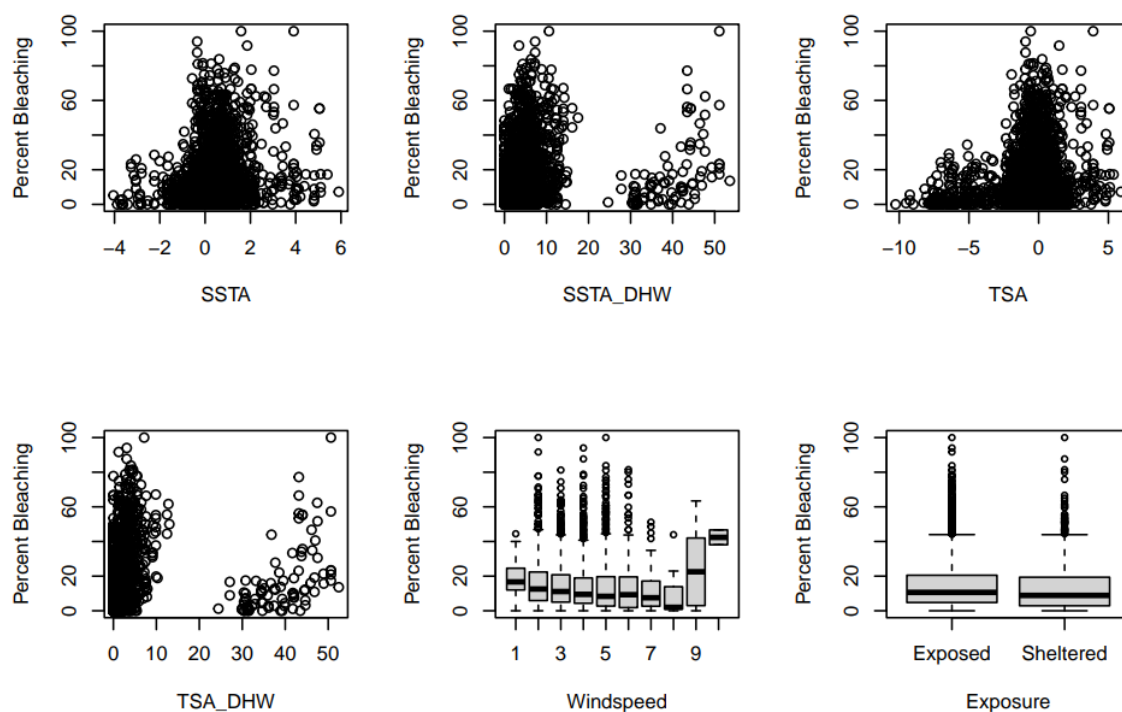


Figure 5 (below): Densities of predictor variables

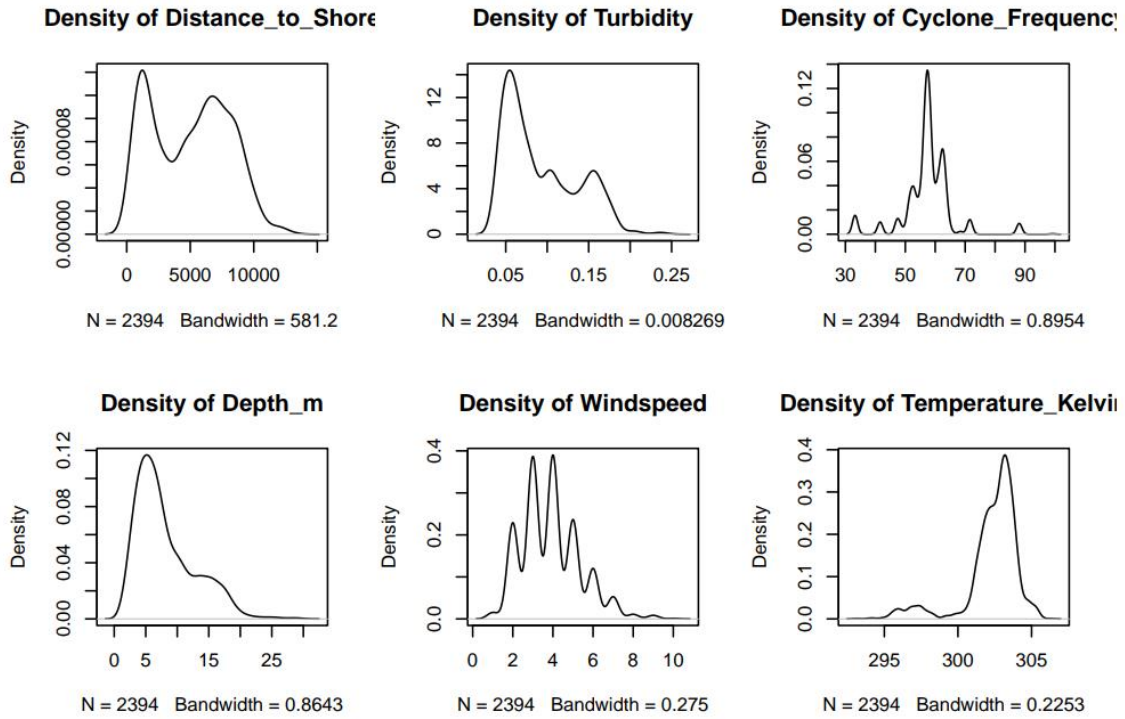


Figure 6 (below): Densities of predictor variables

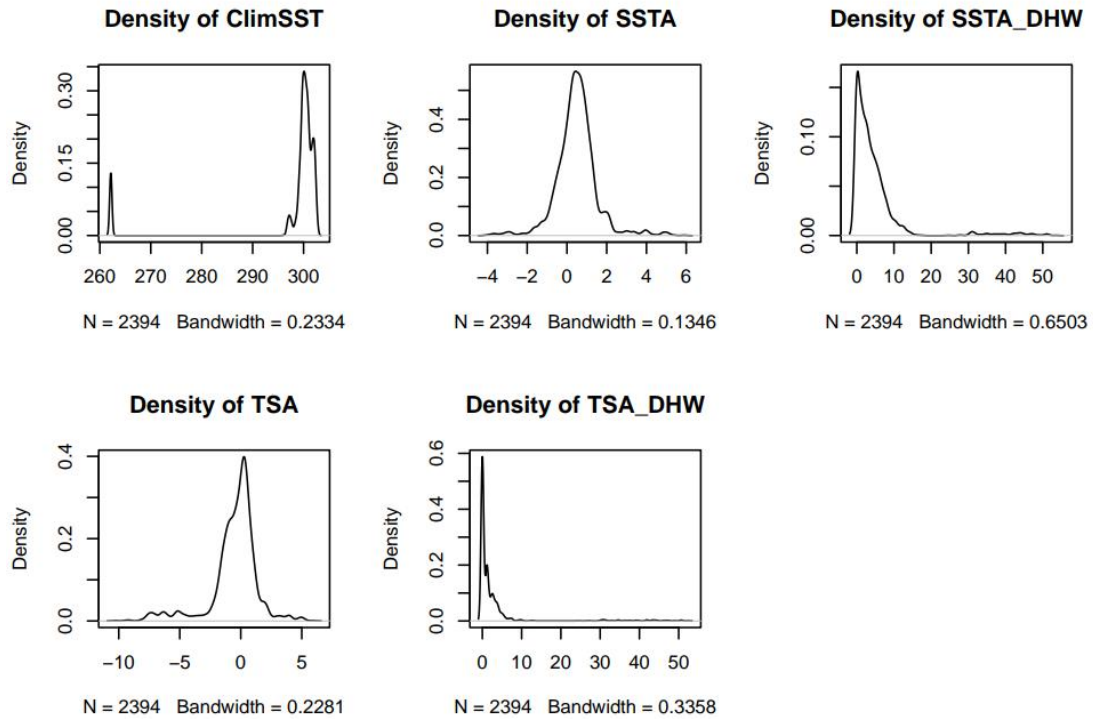


Figure 7 (below): Graph of test MSE values for both MLR models

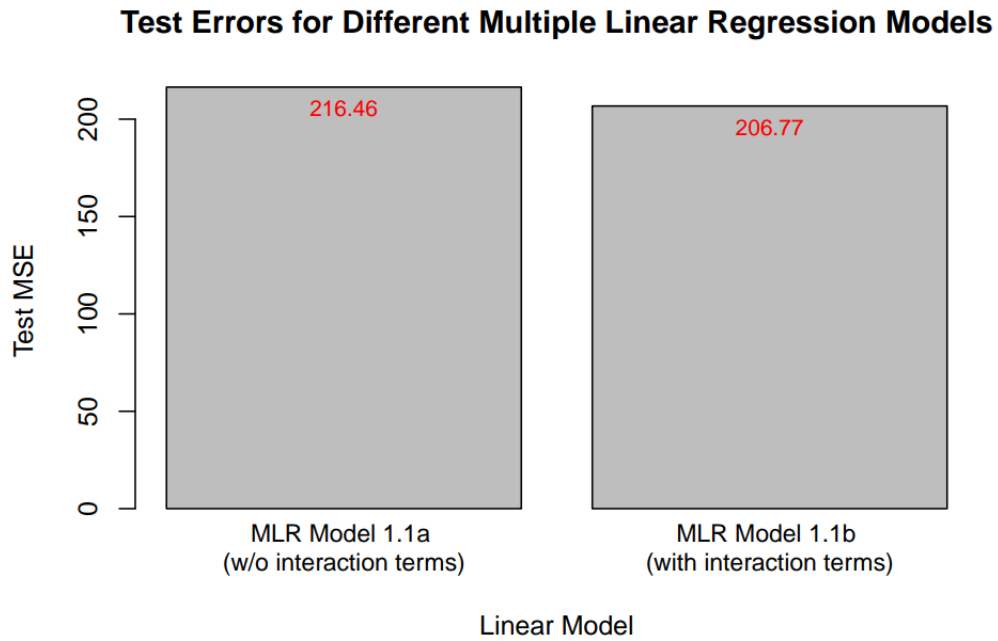


Figure 8 (below): Graph of test MSE values for all linear models

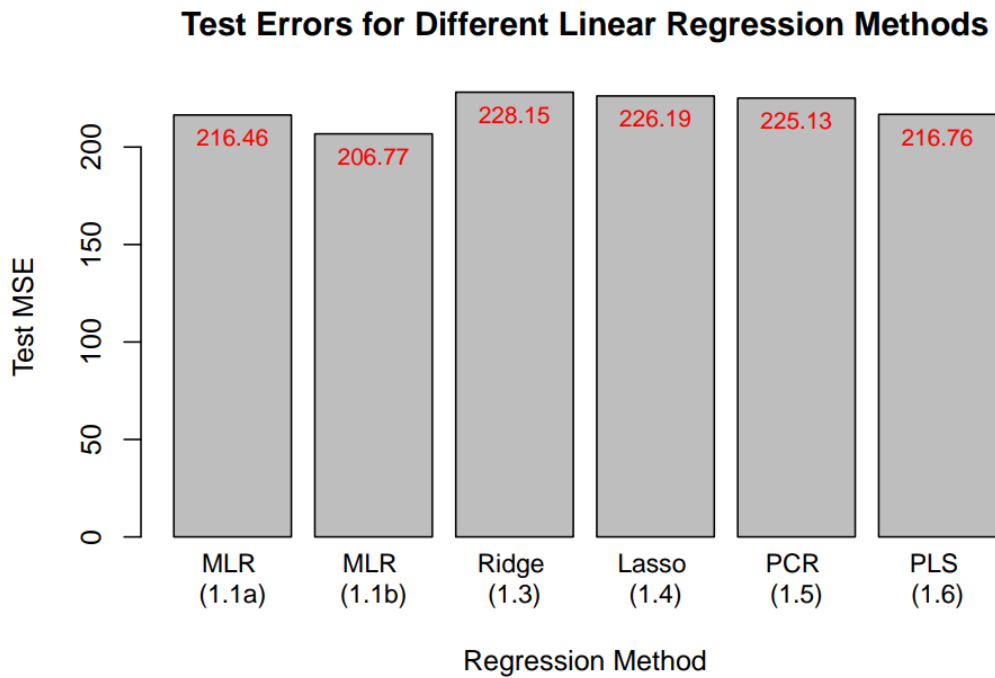


Figure 9 (below): Table of test MSE and RMSE values for all linear models

	MSE	RMSE
MLR (w/o interaction)	216.4616	14.71264
MLR (with interaction)	206.7676	14.37942
Ridge	228.1491	15.10460
LASSO	226.1912	15.03966
PCR	225.1290	15.00430
PLS	216.7603	14.72278

Figure 10 (below): Graph of test MSE values for all nonparametric models

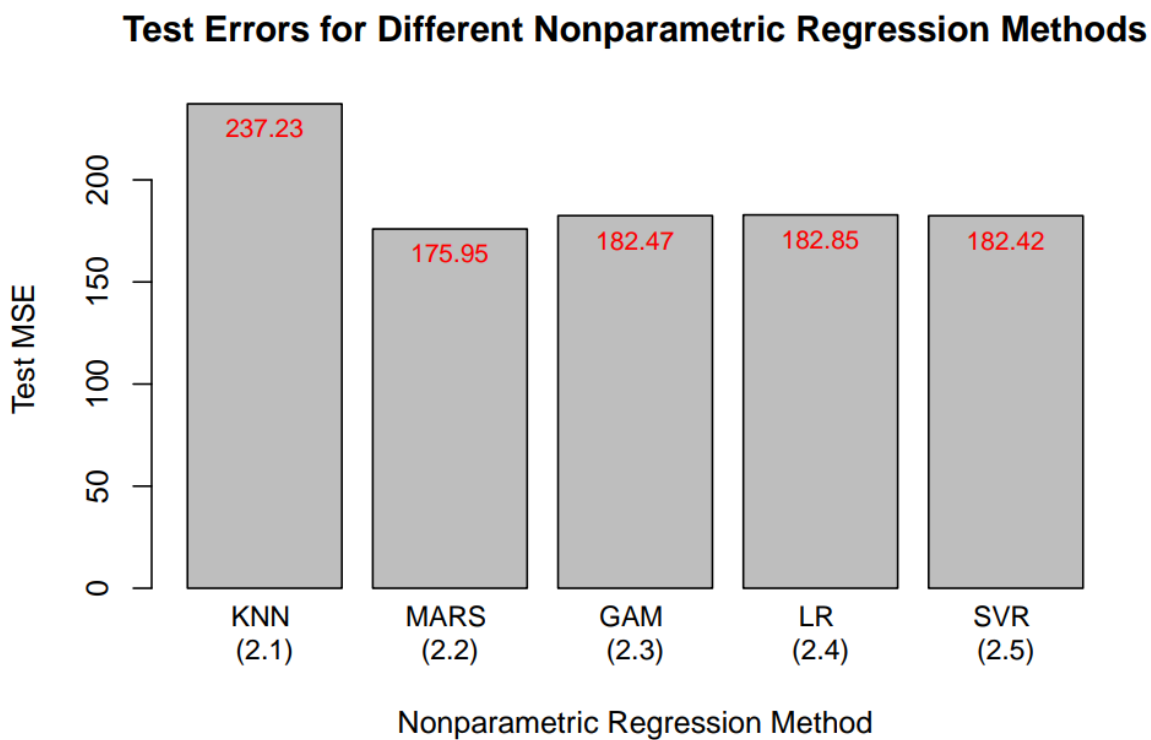


Figure 11 (below): Table of test MSE and RMSE values for all nonparametric models

	MSE	RMSE
KNN	237.2307	15.40230
MARS	175.9543	13.26478
GAM	182.4715	13.50820
LR	182.8450	13.52202
SVR	182.4174	13.50620

Figure 12 (below): Graph of test MSE values for all regression tree models

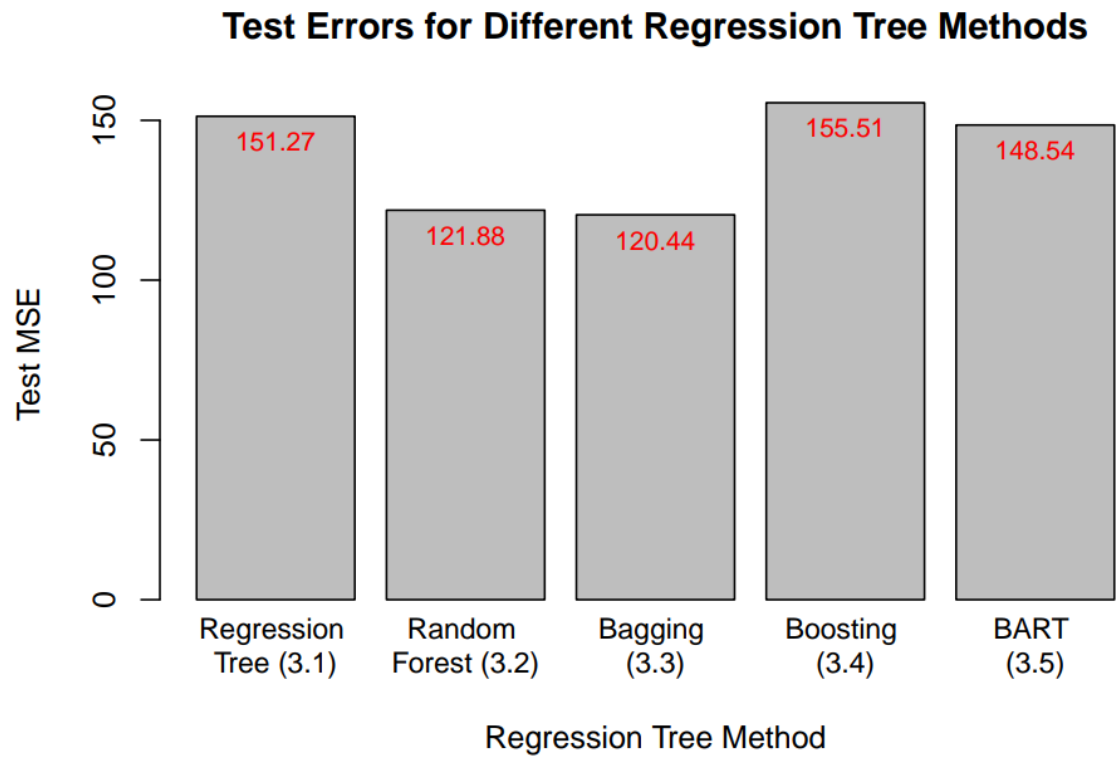


Figure 13 (below): Table of test MSE and RMSE values for all regression tree models

	MSE	RMSE
Regression Tree	151.2734	12.29933
Random Forest	121.8815	11.04000
Bagging	120.4397	10.97450
Boosting	155.5093	12.47034
BART	148.5438	12.18785

Figure 14 (below): Graph of test MSE values for all stacked models

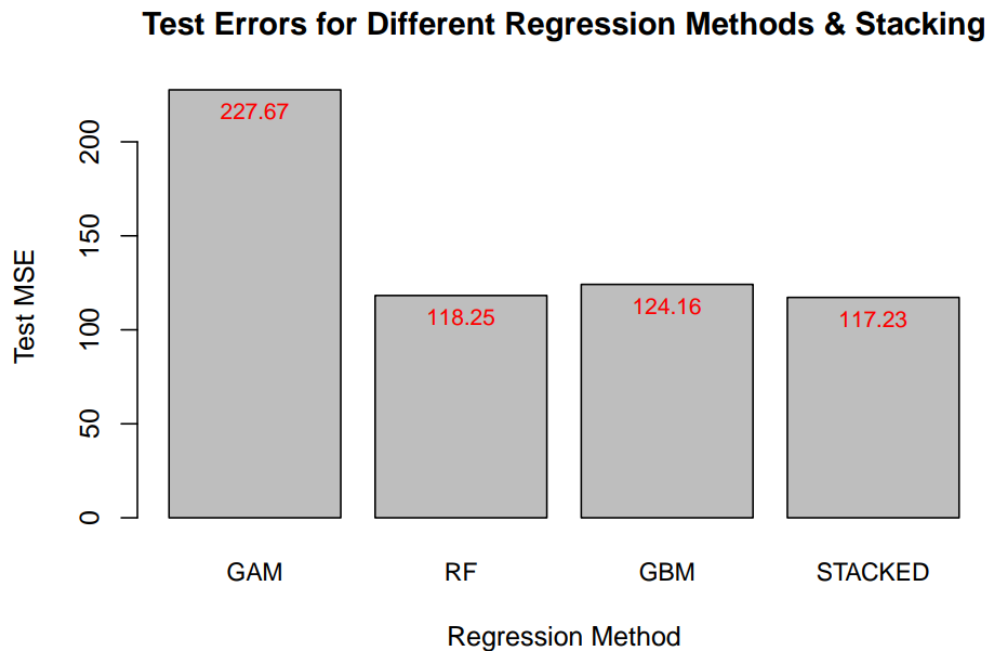


Figure 15 (below): Table of test MSE and RMSE values for all stacked models

	MSE	RMSE
GAM	227.6656	15.08859
RF	118.2482	10.87420
GBM	124.1598	11.14270
STACKED	117.2267	10.82713

Figure 16 (below): Evaluation metrics for MLR model with no interaction terms

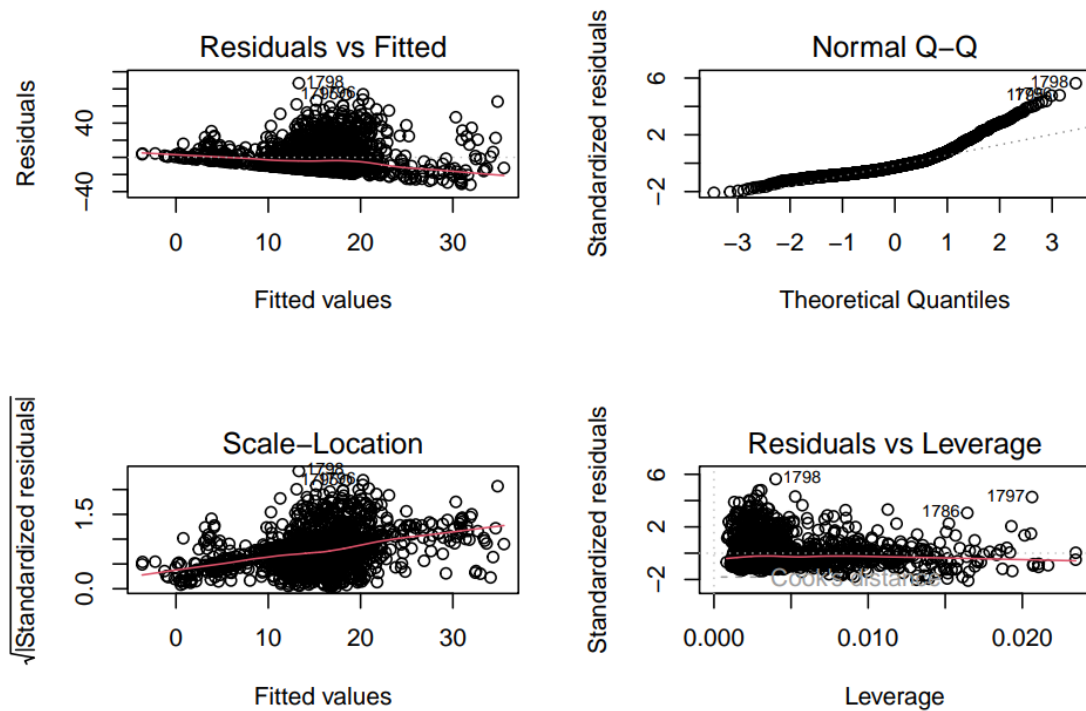


Figure 17 (below): Evaluation metrics for MLR model with interaction terms

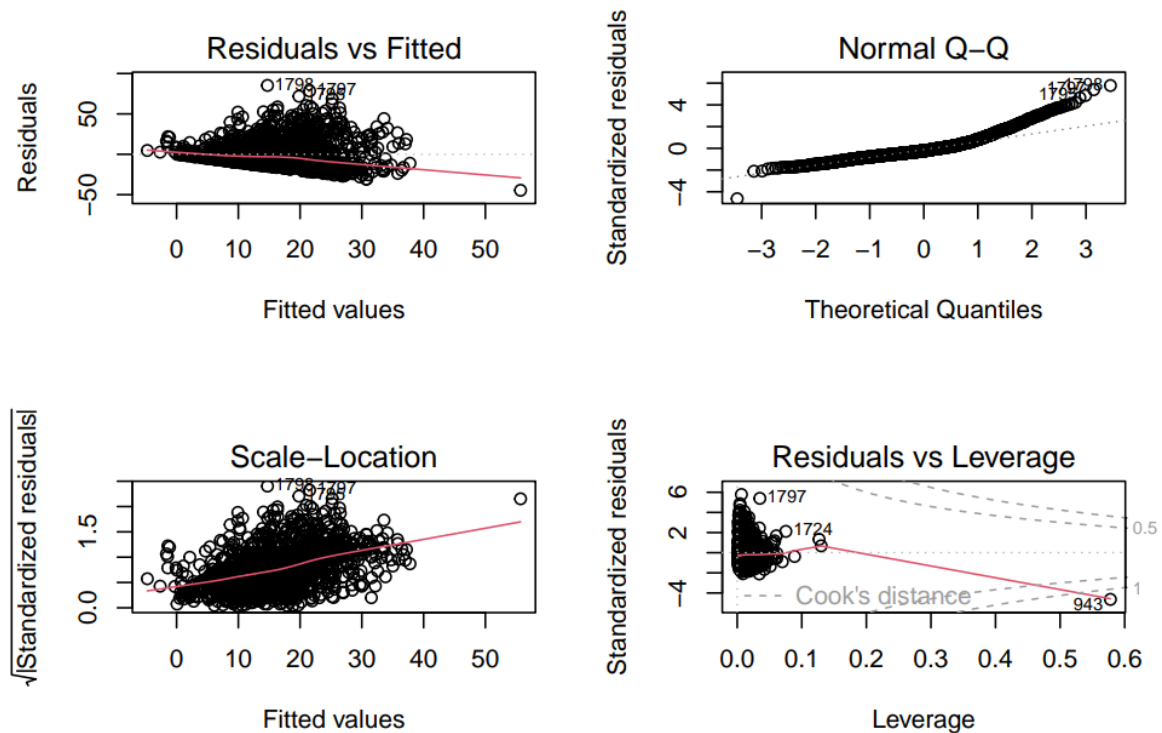


Figure 18 (below): Lambda values for ridge regression

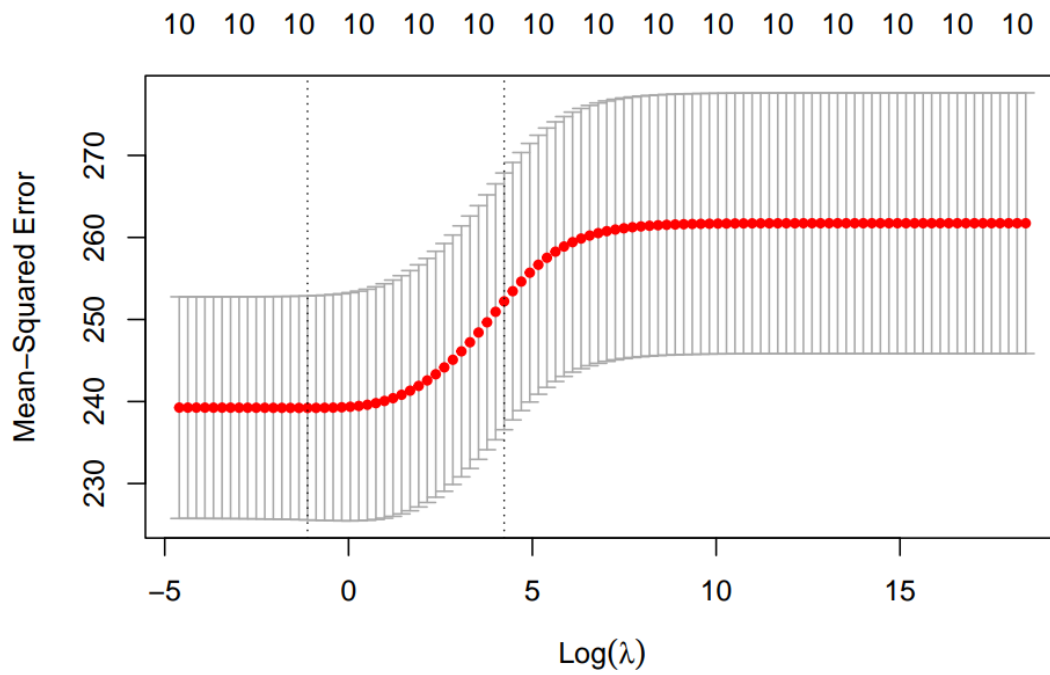


Figure 19 (below): Lambda values for LASSO regression

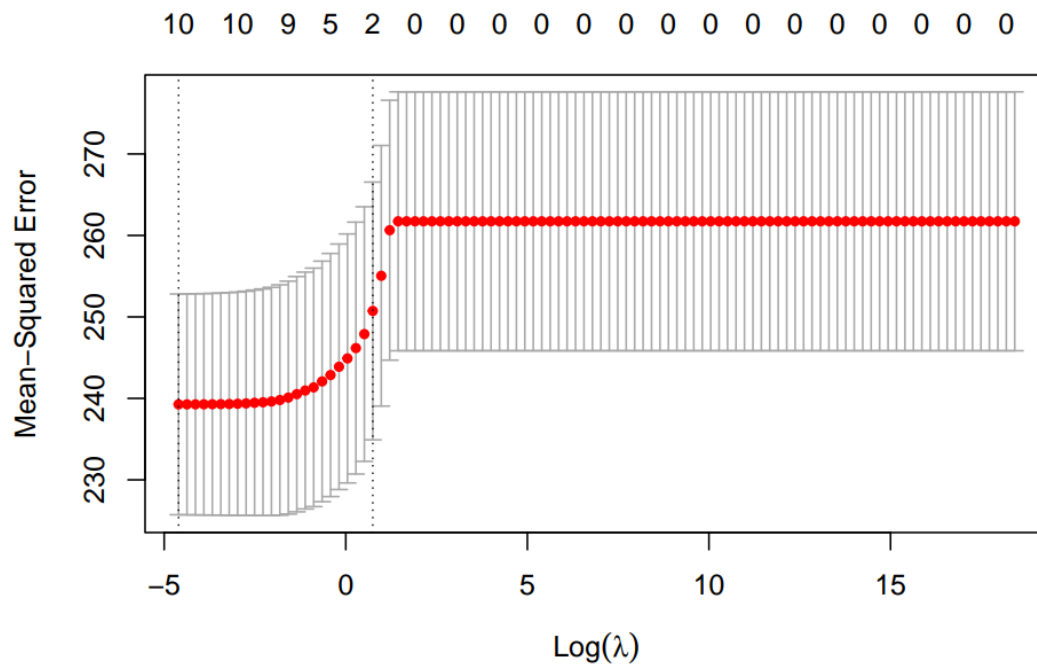


Figure 20 (below): K values for K-nearest neighbors

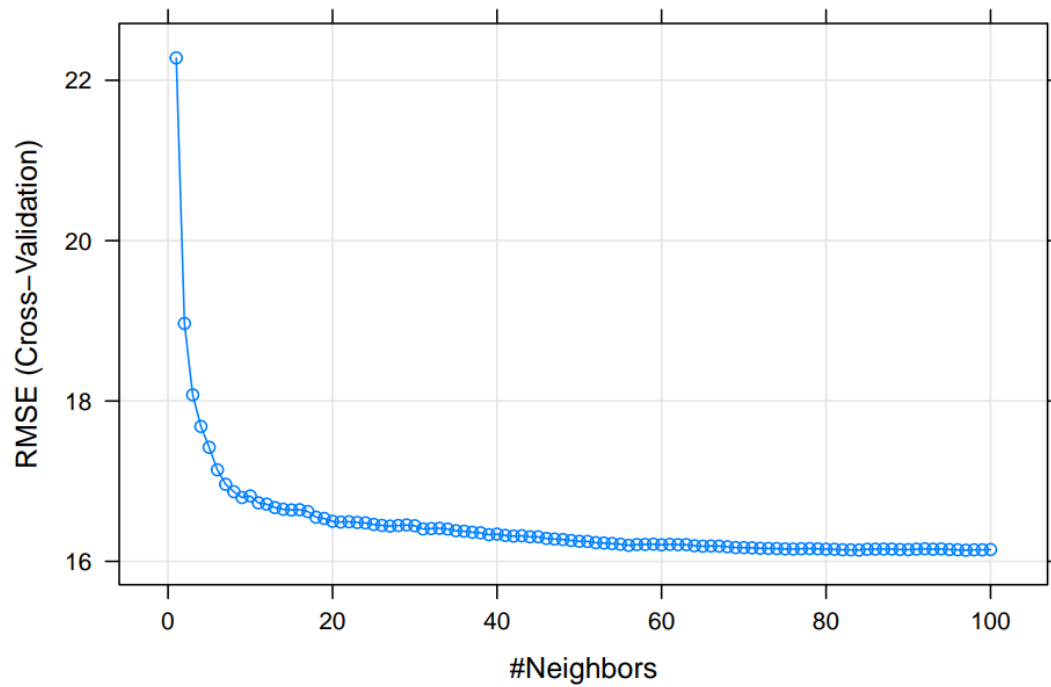


Figure 21 (below): MARS graph

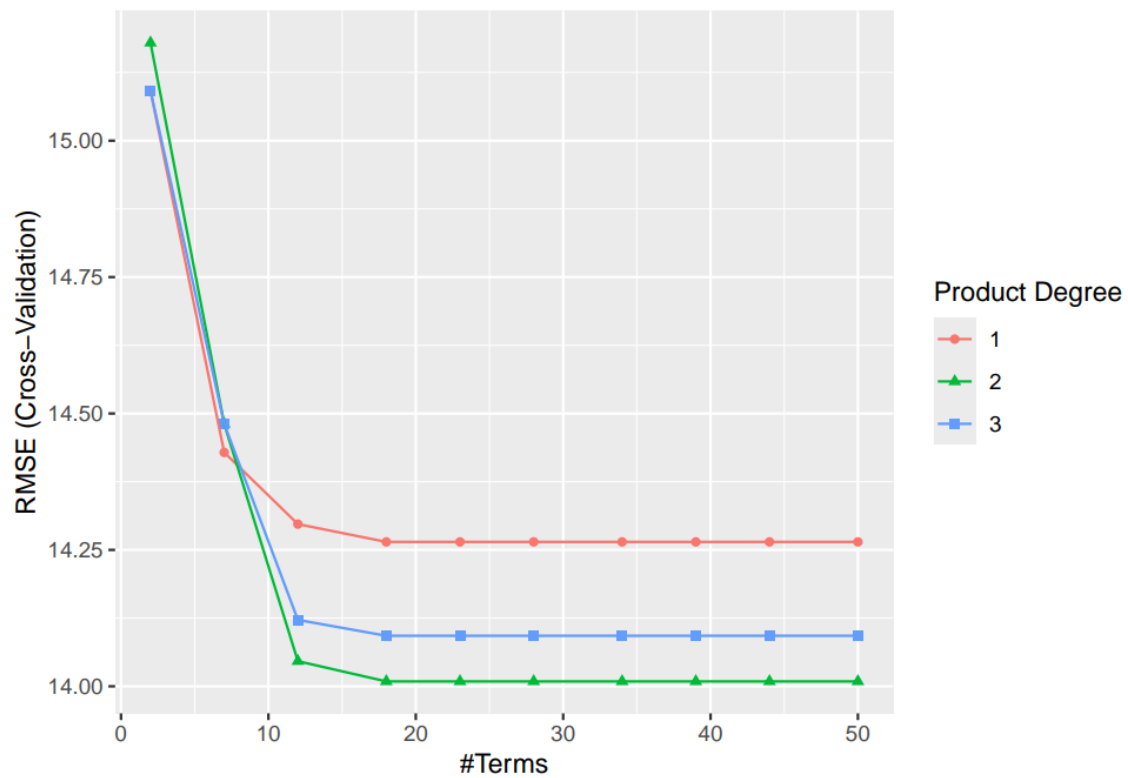


Figure 22 (below): GAM model

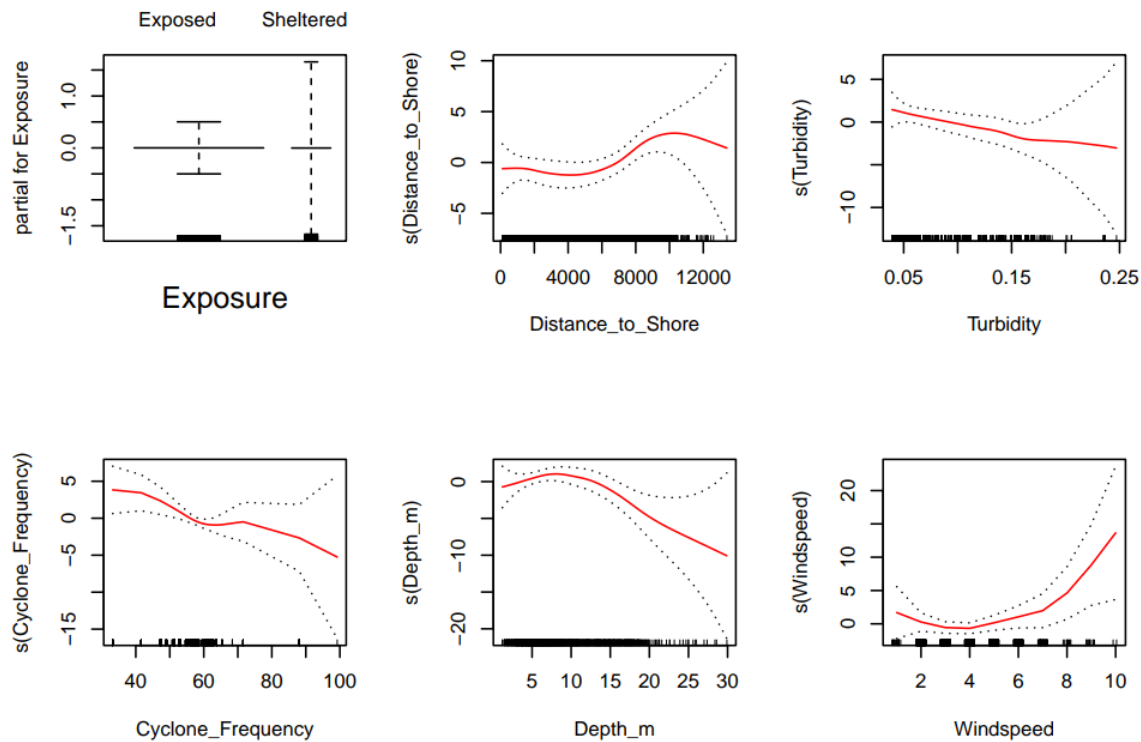


Figure 23 (below): GAM model

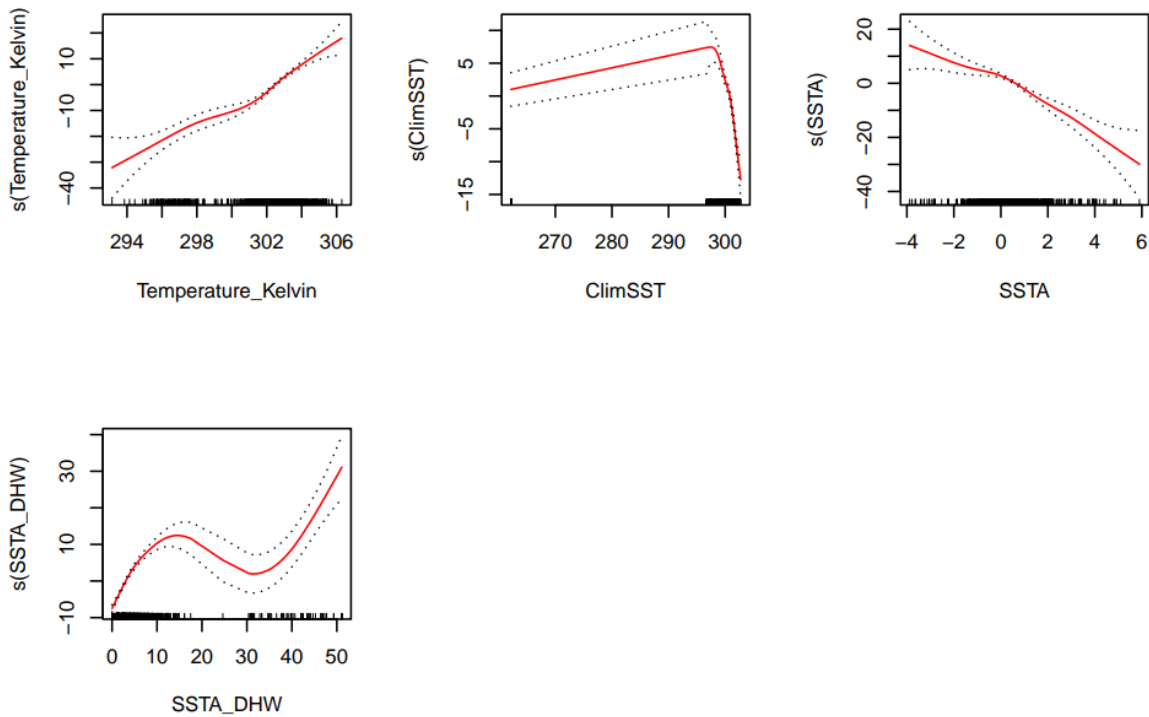


Figure 24 (below): Standard regression tree model

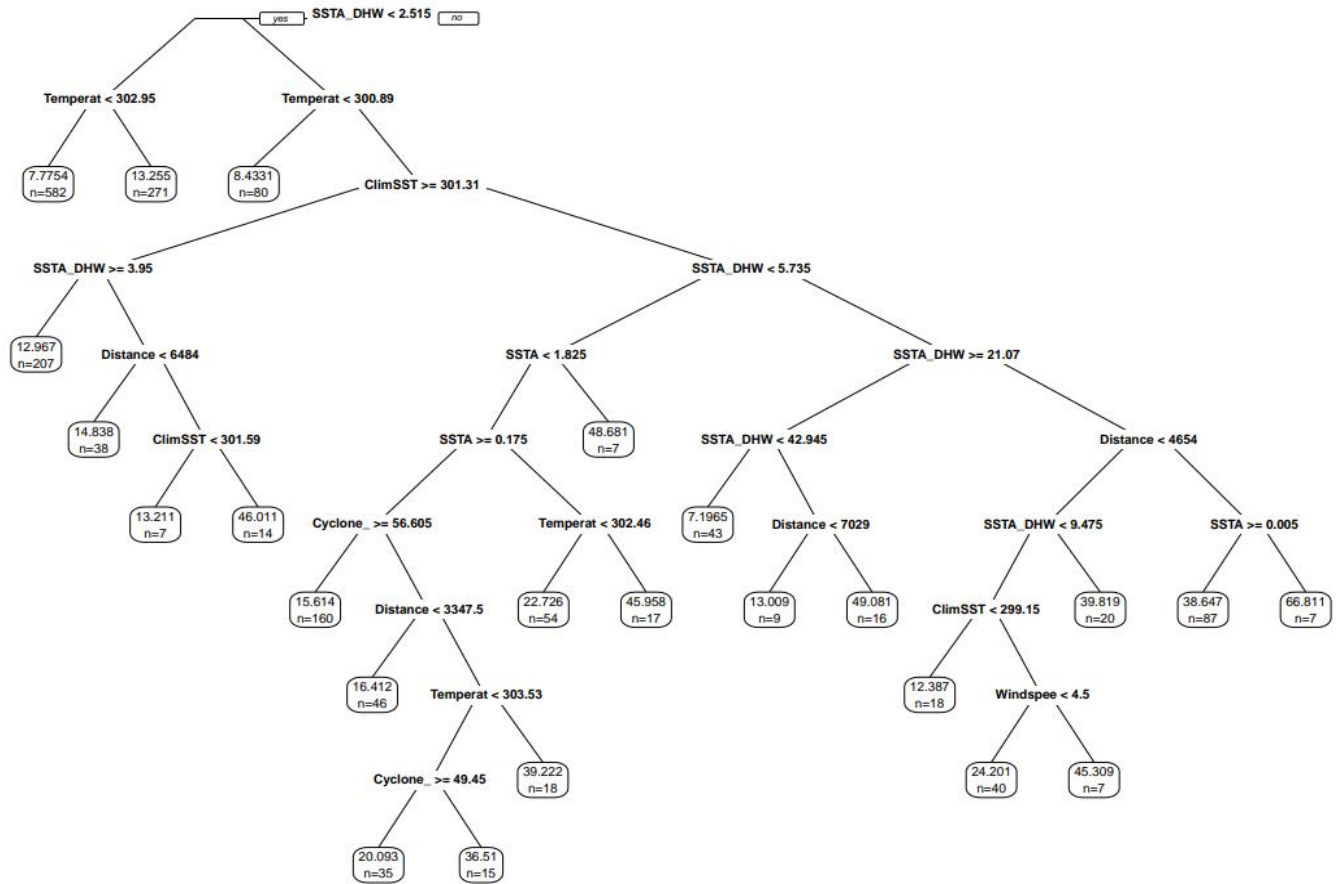


Figure 25 (below): Random Forest model, number of trees

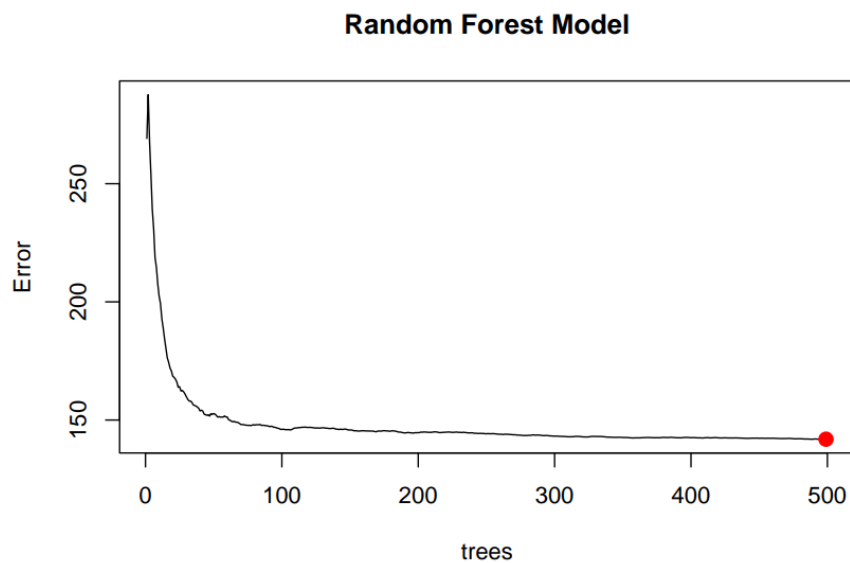


Figure 26 (below): Random Forest model, variable importance

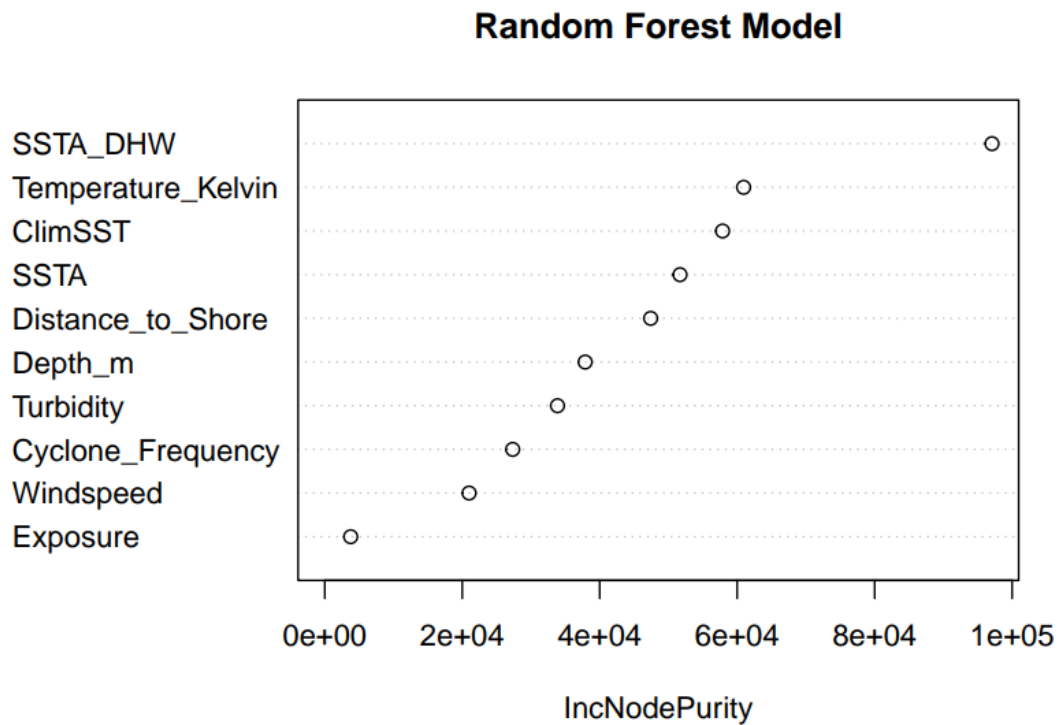


Figure 27 (below): Bagged tree model, variable importance

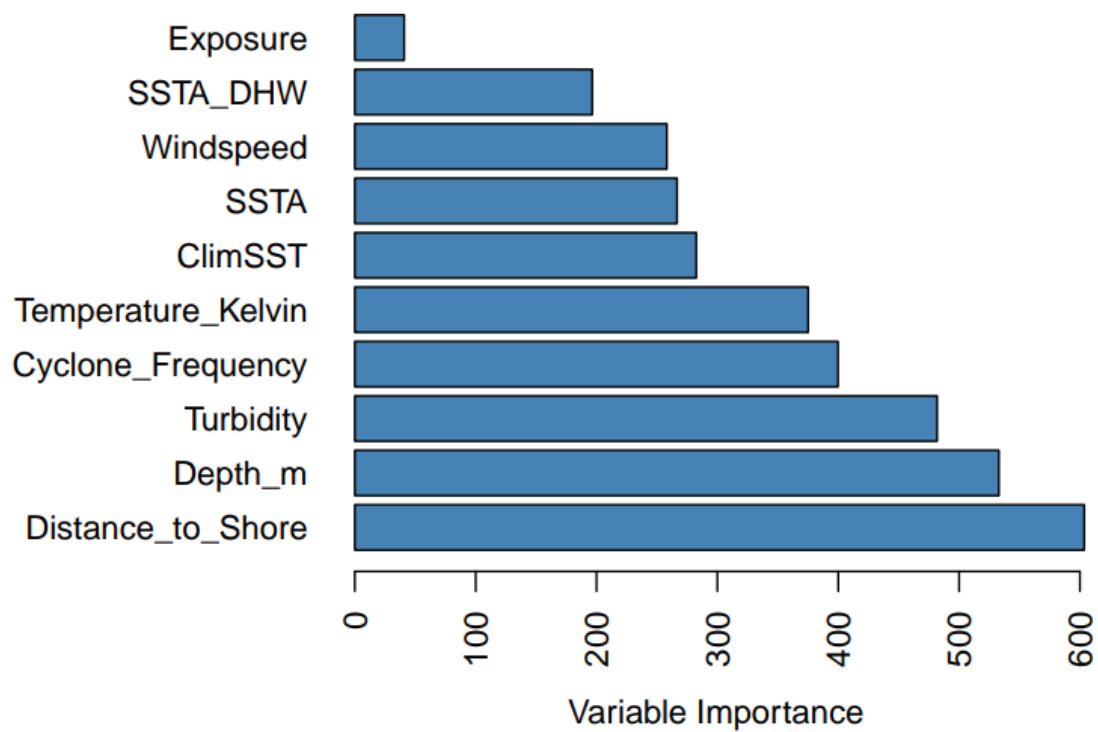


Figure 28 (below): Boosted tree model, variable relative influence (graph)

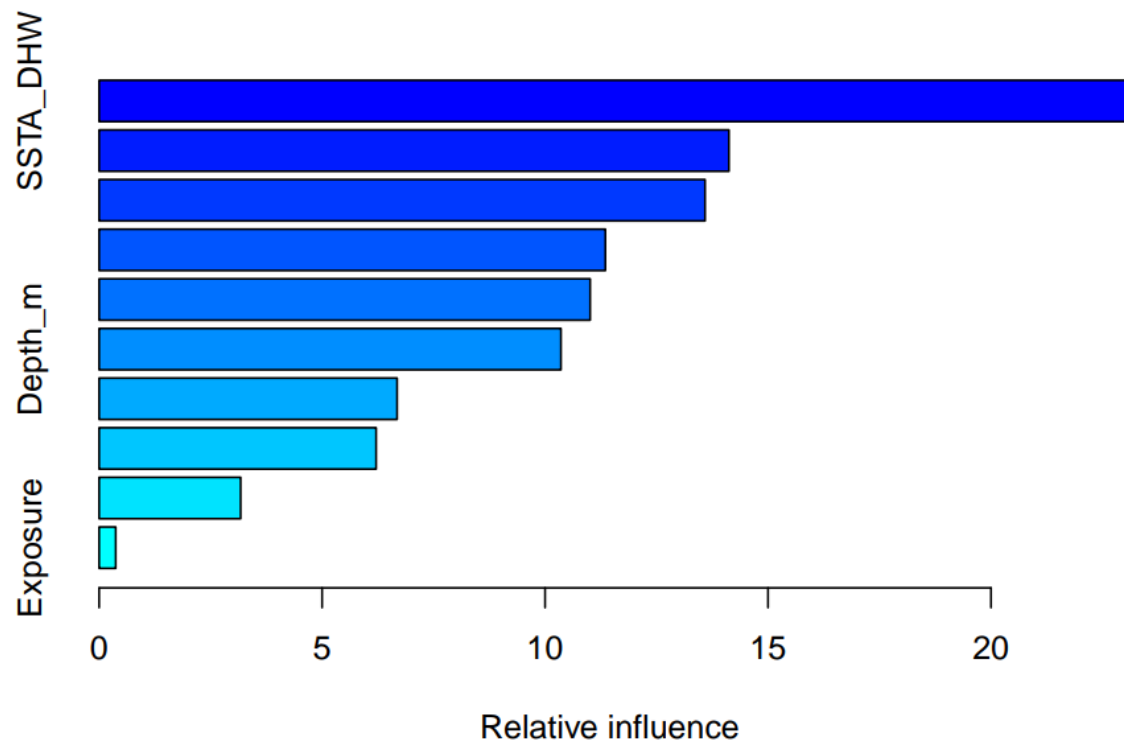


Figure 29 (below): Boosted tree model, variable relative influence (table)

(Intended to show reader what order the variables are in, in reference to Figure 17)

	var <chr>	rel.inf <dbl>
SSTA_DHW	SSTA_DHW	23.1307418
Distance_to_Shore	Distance_to_Shore	14.1229549
ClimSST	ClimSST	13.5877954
SSTA	SSTA	11.3529763
Temperature_Kelvin	Temperature_Kelvin	11.0110495
Depth_m	Depth_m	10.3561369
Turbidity	Turbidity	6.6817838
Cyclone_Frequency	Cyclone_Frequency	6.2114567
Windspeed	Windspeed	3.1731208
Exposure	Exposure	0.3719837

1-10 of 10 rows

Figure 30 (below): Correlations of predictions from stacked model

